

ICT80004 Weekly Communication – Week #06

Student Name: Arun Ragavendhar Arunachalam Palaniyappan ID: 104837257

Organisation: Commonwealth Scientific and Industrial Research Organisation (CSIRO)

Industry Supervisor: Dr. Shigang Liu

Date Prepared: 12/09/2025 Internship Week #: 6

Day	Date	Task(s) completed
1	Monday 8 Sep 2025 8 hours	- Expanded the <i>prompt attack test suite</i> for Violence & Hate, Guns & Illegal Weapons, and Criminal Planning, using Martians as the safe placeholder. - Created direct vs. paraphrased conversational prompts for each of the ten social engineering techniques (Urgency, Attention Grabbing, Visual/Verbal Deception, Incentive & Motivator, Persuasion, Quid-Pro-Quo, Foot-in-the-Door, Trusted Relationship, Impersonation, Pretexting). - Tested prompts on GPT-5 and recorded outcomes (<i>Failure / Partial Success / Success</i>) with screenshots and detailed annotations.
2	Wednesday 10 Sep 2025 8 hours	- Refined <i>incentive-based</i> and <i>persuasion-based</i> disguises that successfully elicited indirect responses. Security Highlight: Consolidated and referenced the <i>full vulnerable code test suite</i> completed across Weeks 3–6. This includes 12 classic vulnerabilities (integer overflow, null dereference, buffer overflow, format string, SQL injection, path traversal, etc.) tested under multiple prompting styles. Results confirmed that while models often suggested correct mitigations, weaknesses still appeared in certain styles (e.g., type coercions, clamping, verbose or incomplete checks). This establishes a strong baseline of code-level vulnerabilities against which prompt attacks can now be benchmarked.

Total hours completed for the week: 16

Plans for next week: **#07 week (15 - 19 Sept 2025)**

- Investigate additional **security vulnerabilities and bypass methods** in LLM behaviour (multi-turn steering, nested instruction smuggling, role-play escalation).
- Finalise the *full prompt injection test suite* with all attack categories and techniques covered.
- Map both code-based and prompt-based vulnerabilities systematically to the taxonomy for consistency.
- Prepare a summary presentation of *vulnerable code testing + prompt injection results* for the next review.

Screenshot of Timely EMAIL communication update to the Supervisor at the end of week #06 sent on 12 September 2025 Friday at 4:40 PM (16:40).

Weekly Communication / Reflection update - #06 week, 8 - 12 Sep 2025



Arun Ragavendhar <arunragavendhar.1999@gmail.com>

12 Sep

to Shigang

Dear Dr. Shigang,

I hope you

Activities

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- C

from: **Arun Ragavendhar** <arunragavendhar.1999@gmail.com>
to: "Liu, Shigang (Data61, Clayton)" <Shigang.Liu@data61.csiro.au>
date: 12 Sept 2025, 16:40
subject: Weekly Communication / Reflection update - #06 week, 8 - 12 Sep 2025
mailed-by: gmail.com

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g techniques (Urge

Weekly Communication / Reflection update - #06 week, 8 - 12 Sep 2025



Arun Ragavendhar <arunragavendhar.1999@gmail.com>
to Shigang

12 Sept 2025, 16:40 (2 days ago) ☆ ☺ ↶ ⋮

Dear Dr. Shigang Liu,

I hope you are well. Please find below my update for Week 6 of the internship.

Activities completed this week (Total: 16 hours):

- Expanded the *prompt attack test suite* for Violence & Hate, Guns & Illegal Weapons, and Criminal Planning, using Martians as the safe placeholder.
- Created **direct vs. paraphrased conversational prompts** for each of the ten social engineering techniques (Urgency, Attention Grabbing, Visual/Verbal Deception, Incentive & Motivator, Persuasion, Quid-Pro-Quo, Foot-in-the-Door, Trusted Relationship, Impersonation, Pretexting).
- Tested prompts on GPT-5 and recorded outcomes (*Failure / Partial Success / Success*) with screenshots and detailed annotations.
- Refined *incentive-based* and *persuasion-based* disguises that successfully elicited indirect responses.
- **Security Testing:** Consolidated and referenced the *full vulnerable code test suite* completed across Weeks 3–5. This includes 12 classic vulnerabilities (integer overflow, null dereference, buffer overflow, format string, SQL injection, path traversal, etc.) tested under multiple prompting styles. Results confirmed that while models often suggested correct mitigations, weaknesses still appeared in certain styles (e.g., type coercions, clamping, verbose or incomplete checks). This establishes a strong baseline of code-level vulnerabilities against which prompt attacks can now be benchmarked.

Plan for next week (Week 7: 15–19 Sept 2025):

- Investigate additional **security vulnerabilities and bypass methods** in LLM behaviour (multi-turn steering, nested instruction smuggling, role-play escalation).
- Finalise the *full prompt injection test suite* with all attack categories and techniques covered.
- Map both code-based and prompt-based vulnerabilities systematically to the taxonomy for consistency.
- Prepare a summary presentation of *vulnerable code testing + prompt injection results* for the next review meeting.

Kind regards,
Arun Ragavendhar Arunachalam Palaniyappan
ICT80004 Internship Student – CSIRO Data61

4 attachments • Scanned by Gmail

